

Prewarming

Warm chosen nodes *before* jobs arrive.

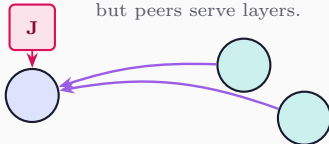


fewer cold hits: $c/N \downarrow$

Changes *whether* a job hits a cold node.
Reduces the number of affected jobs.

Cluster-local mirror

A cold hit still happens,
but peers serve layers.



cheaper pulls: $p_I \downarrow$

Changes *how costly* a cold hit is. Re-
duces p_I , not the hit probability.

Complementary: prewarming avoids cold hits; a mirror makes the remaining cold hits cheaper.